Trust-Anchored Federated Learning for Robust Specification-AI Hybrid Detection of False Base Station Attacks*

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Abstract

False Base Stations (FBS) remain a persistent threat to 4G LTE and 5G networks, enabling identity catching, service downgrading, and denial-of-service attacks by exploiting the asymmetric authentication model of cellular protocols. Specification-based detection systems derive formally verified behavior rules from 3GPP normative clauses and achieve complete coverage on known attack classes, while recent hybrid extensions augment these static rules with LSTM-based binary classifiers trained across user equipment (UE) via Federated Learning (FL) to capture multi-step temporal attack sequences. However, existing hybrid architectures assume that all participating UEs are honest, leaving the FL aggregation step exposed to compromised devices that submit poisoned model updates. This paper proposes a trust-anchored aggregation mechanism that closes this deployability gap. The aggregator maintains a small set of formally verified clean LTE captures as a root reference dataset and, in each FL round, computes its own honest update against this reference; each client's submitted update is then scored by directional consistency with the reference and rescaled to a bounded magnitude before contributing to the global model. A second behavioral trust anchor cross-checks each client's local LSTM predictions against the underlying formally verified behavior rules, demoting clients whose models persistently disagree with normatively-grounded ground truth regardless of how their weight-space updates appear. The combined mechanism is designed to defend against backdoor injection, untargeted gradient poisoning, scaled-magnitude attacks, and Sybil collusion without requiring prior knowledge of the Byzantine fraction. This work proposes a threat model for adversarial federated FBS detection that formalizes the deployability gap in prior hybrid architectures, a trust-anchored aggregation rule that fuses weight-space directional consistency with specification-rule behavioral verification into a unified per-client trust score, and a proposed evaluation framework targeting on-device deployment via a rooted Android UE connected to an Open5GS core with srsRAN eNodeB, with simulated federation peers and varying compromise fractions.

*Proceedings of The 2026 IFIP WG 8.4 International Symposium on E-Business Information Systems Evolution (EBISION 2026), Article No. P-19, July 8–10, 2026, Seoul, South Korea.
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